Autonomous UAV System for Precision Crop Monitoring and Terrain-Adaptive Path Planning

Matías Dolgay
School of Engineering
UdelaR
Montevideo, Uruguay
0009-0005-3694-4016

Romina Sosa
School of Engineering
UdelaR
Montevideo, Uruguay
0009-0001-9327-3053

Facundo Benavides
School of Engineering
UdelaR
Montevideo, Uruguay
0000-0002-0651-5246

***Abstract*—In modern precision agriculture, timely weed detection is critical for maintaining crop yield and sustainability. However, traditional crop monitoring methods are labour-intensive and prone to human error. To address these challenges, this paper presents an autonomous Unmanned Aerial Vehicle (UAV) system designed for high-resolution crop inspection and optimised path planning. The proposed system integrates the ROS 2 framework and the PX4 flight stack in a multi-stage workflow. First, a reconnaissance flight provides imagery for generating a georeferenced orthomosaic and elevation map using OpenDroneMap. Digital image processing techniques, including morphological operations and contour analysis, are then used to detect crop rows. A Generalised Voronoi Diagram (GVD) is constructed from the detected rows to represent the field layout and generate navigation paths through the inter-row spaces. The system was evaluated in a robotic simulation environment using different field geometries to assess its ability to generate suitable trajectories for autonomous inter-row navigation and agricultural image acquisition.**

***Index Terms*—Precision Agriculture, Autonomous UAVs, Path Planning, Generalised Voronoi Diagram, ROS 2, Computer Vision, Crop Row Detection, Weed Monitoring.**

I. Introduction

In modern agriculture, efficient crop monitoring is essential for sustainable and productive farming. Access to accurate information about crop conditions and potential anomalies enables timely decision-making. However, traditional monitoring methods based on manual field inspection are labour-intensive, time-consuming, and prone to human error [1]. This is particularly relevant for weed management, as weeds compete with crops for essential resources such as water, sunlight, and nutrients, potentially reducing crop yield and quality. Their effective management requires frequent field monitoring, which further increases the operational workload [2].

Within a broader research project on automated weed detection, this work focuses on developing an autonomous flight and path-planning system for agricultural image acquisition. Unmanned Aerial Vehicles (UAVs) offer a flexible solution for this task, enabling aerial data collection over agricultural areas, including locations that may be difficult to access on foot.

The main objective of this study is to generate flight trajectories through the spaces between rows of fruit crops, providing a basis for the subsequent acquisition of images for automated weed detection. The proposed system focuses on autonomous navigation and trajectory planning, rather than on weed detection itself.

To achieve this objective, we designed a multi-stage system. First, an operator performs an initial manual flight to establish a reference route for autonomous field reconnaissance. The resulting flight is used to generate the aerial data required to create a detailed map of the field.

Next, image-processing techniques are applied to the generated map to identify the crop rows. The detected rows are then used to generate flight trajectories through the inter-row spaces, with the aim of supporting systematic image acquisition for subsequent weed detection.

The proposed system is evaluated in a robotic simulation environment, where the generated flight trajectories are tested under controlled conditions and across different field configurations. This evaluation provides an initial assessment of the proposed approach before its implementation on a physical UAV.

The remainder of this paper is structured into five sections. Section I (this introduction) establishes the objectives, problem statement, and scope of the work. Section II presents a literature review, addressing relevant studies and exploring the use of UAVs in agriculture with an emphasis on navigation strategies. Section III details the technical development of the system, including UAV navigation and crop map creation and processing. Section IV describes the experimental setup used to validate the system, while Section V provides the conclusions and outlines potential directions for future work.

II. Background and Related Work

Precision agriculture increasingly relies on UAVs because they enable the acquisition of high-resolution aerial imagery and flexible monitoring of large cultivated areas.

In this context, weed detection is particularly relevant, as weeds compete with crops for resources such as water, nutrients, and land, thereby affecting productivity and motivating automated monitoring and site-specific management strategies.

Recent trends in Site-Specific Weed Management indicate that the transition from broad spraying to localised intervention relies heavily on the quality of aerial data acquisition [3].

While modern models have demonstrated high accuracy for real-time crop and weed classification, their performance is often limited by operational challenges in data acquisition. Consequently, specialised flight stacks based on Robot Operating System 2 (ROS 2) and PX4 have become essential for standardising autonomous collection in complex environments.

Previous research has explored the use of UAVs in precision agriculture for tasks such as crop monitoring, weed detection, and autonomous navigation. Several papers focus on applying computer vision techniques to identify weeds in aerial images. For instance, [1] presents a monocular vision-based approach using a fused-YOLO architecture to detect multiple weed types and estimate their relative position. Similarly, [4] evaluates the performance of the YOLOv5 model for crop and weed classification using UAV images, demonstrating high accuracy and real-time applicability. Modern systematic reviews further highlight a pivot toward Transformer-based architectures to distinguish weeds from morphologically similar crops [5]. However, the effectiveness of these advanced artificial intelligence (AI) models is highly dependent on the "angle of attack", the specific perspective from which the UAV captures the inter-row space, necessitating more deliberate navigation strategies.

Other works highlight the potential of UAVs for weed control and monitoring in agricultural environments. [6] analyses the use of UAV-based remote sensing for integrated weed management in small-scale farming systems. Emphasising the role of aerial images in mapping weed distribution, [7] reviews UAV applications for herbicide spraying and weed detection, reporting improvements in coverage efficiency and reduced environmental impact compared to traditional methods.

In addition to weed detection, autonomous navigation and trajectory planning for UAV-based agricultural systems are addressed in [8], which proposes trajectory-planning methods using active sensors to improve mapping accuracy in agricultural scenarios. Complementarily, [9] presents an autonomous ground vehicle platform for mapping and spraying tasks, demonstrating the integration of sensing and automated operation in agricultural environments.

Additionally, open-source platforms such as CoFly provide support for AI technologies to automate the analysis of agricultural data [10].

UAV navigation strategies can be categorised into adaptive and predefined approaches. Adaptive navigation dynamically adjusts the UAV trajectory based on sensor data and environmental conditions, whereas predefined navigation relies on previously planned Global Positioning System (GPS) waypoints to simply execute the mission.

Although adaptive navigation can adjust the trajectory online in response to sensed environmental changes, it also requires greater onboard processing for simultaneous perception, obstacle handling and path generation. In contrast, predefined navigation reduces computational demand during flight by relying on previously generated waypoints [8].

In this approach, the priority is to dedicate most of the processing to identifying crop rows, rather than on UAV navigation [11]. This approach enables a more efficient allocation of system resources by ensuring that the UAV prioritises its primary function (high-precision monitoring and analysis of crop conditions) while preventing the navigation subsystem from consuming excessive computational capacity.

III. Autonomous UAV-based Solution

The proposed solution comprises an autonomous UAV-based system for crop inspection and navigation between rows in fruit plantations. The system targets crops arranged in approximately parallel rows with regular spacing between them, with trees typically taller than 1 m. The system was implemented and validated in a simulation environment using ROS 2, PX4, Gazebo, Python, OpenCV, and OpenDroneMap (ODM)[1].

As shown in Fig. 1, the proposed solution consists of a sequential pipeline. First, the area of interest is defined by a manual flight and used to generate a reconnaissance flight plan. During this second flight, the UAV acquires red, green, blue and depth (RGBD) data from the field. Then, the data are processed to obtain a height map, enabling the detection of rows. Based on this information, a second flight plan is generated to guide the UAV autonomously between rows.

A. Area Definition

The process begins with a teleoperated flight used to delimit the area of interest. During this stage, the UAV GPS positions are recorded and later processed to reconstruct the field perimeter. To reduce redundancy while preserving the boundary's geometric structure, the recorded trajectory is simplified using the Ramer–Douglas–Peucker algorithm [12]. The resulting perimeter is then transformed from geographic coordinates into a local North–East–Down (NED) reference frame, using the UAV takeoff position as the origin, to facilitate geometric computations and path planning.

B. Reconnaissance Flight Planning

Using this perimeter, the system generates an initial reconnaissance flight plan composed of parallel trajectories that cover the entire field. To compute these trajectories, the longest side of the perimeter polygon is identified and used as the main orientation of the crop field. Parallel lines are then generated across the polygon in that direction to ensure full coverage of the area. An example of the flight plan is shown in Fig. 2.

[1]https://opendronemap.org/

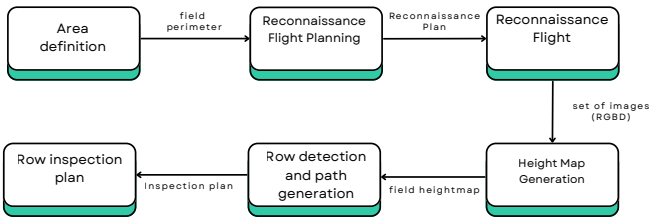


Fig. 1. Overall inspection process pipeline

The spacing between these trajectories is determined based on the UAV altitude, camera field of view, and image acquisition frequency to ensure sufficient overlap between captured images. To avoid abrupt turns at the boundaries of the field, two auxiliary waypoints are generated for each trajectory at a distance δ (1.5 m by default) from the intersections between the parallel lines and the perimeter. These auxiliary points allow the UAV to perform smoother direction changes outside the crop area while maintaining a stable trajectory within the field. The use of these points was motivated by the image quality constraint observed during early tests. Initial reconnaissance paths based directly on perimeter intersections introduced abrupt turns that degraded image quality near the end of each flight line. The value of δ was set empirically: at the cruise speed of 2.0 m/s, smaller distances still produced attitude changes inside the field boundary, while larger values increased flight time without a noticeable improvement. During this reconnaissance flight, the UAV captures georeferenced depth images, which are subsequently used to reconstruct an orthomosaic (height map) of the crop using ODM.

C. Height Map Generation

The captured images are processed to generate a georeferenced height map of the field. For this purpose, the ODM photogrammetry pipeline is used to reconstruct an orthomosaic from the images acquired during the reconnaissance flight. Since ODM expects three-channel RGB images, the captured depth images are converted into RGB format by replicating the depth value across the three colour channels. This transformation preserves the original depth information while making the images compatible with the photogrammetry pipeline. Fig. 3 shows an example of the resulting orthomosaic, which represents the field's height distribution and provides the spatial information needed to identify crop rows.

D. Row Detection and Path Generation

The generated height map enables the identification of crop structures by distinguishing vegetation from ground.

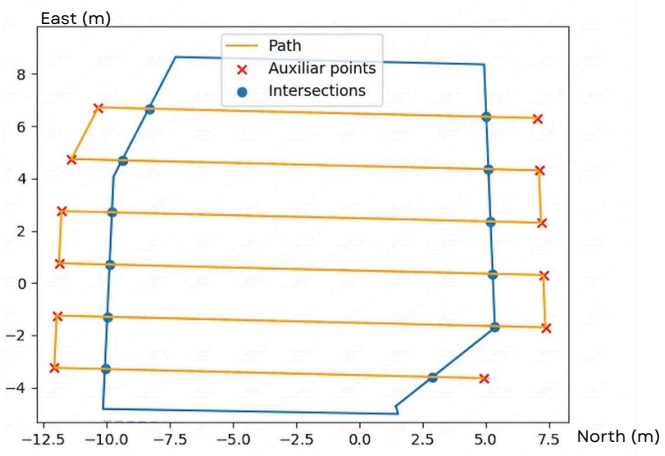


Fig. 2. Reconnaissance flight example. The outer boundary (blue) corresponds to the crop perimeter.

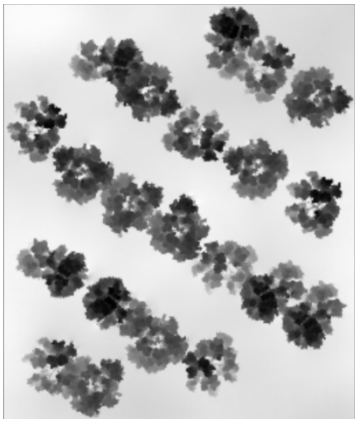


Fig. 3. Height map example. Grey tones denote low-lying areas and black indicates peak heights

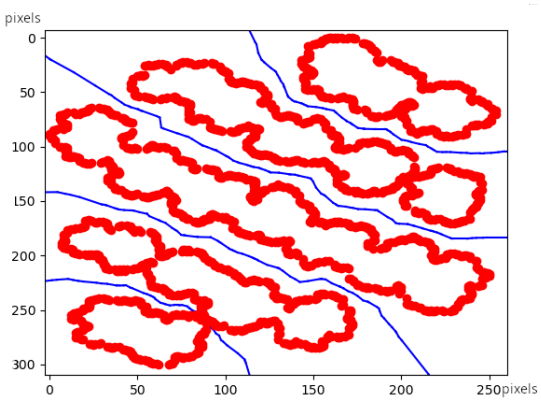


Fig. 4. Row detection (red) and generated navigation paths (blue)

A threshold-based segmentation is applied to separate tree canopies from the soil, followed by contour detection and filtering to identify the crop rows. The detected contours primarily correspond to tree canopies, which may present local irregularities due to variations in foliage density, missing trees, or structural asymmetries.

To determine navigable paths between rows, a Generalised Voronoi Diagram (GVD) is computed using polygons derived from the detected row contours. Unlike standard Voronoi diagrams defined over point seeds, the generalised formulation allows the representation of entire crop rows as geometric regions, thereby better capturing their spatial structure. In our implementation, each row contour is assigned a unique label, which is then propagated to its sampled boundary points. During the Voronoi construction, edges generated between points belonging to the same row are discarded, ensuring that only edges separating different rows are preserved. The resulting Voronoi edges lie approximately equidistant from adjacent rows, thereby defining central corridors that can be used as safe navigation paths for the UAV, as shown in Fig. 4.

E. Row Inspection Flight

From these Voronoi vertices, the system generates a continuous path that traverses the crop rows. To determine the

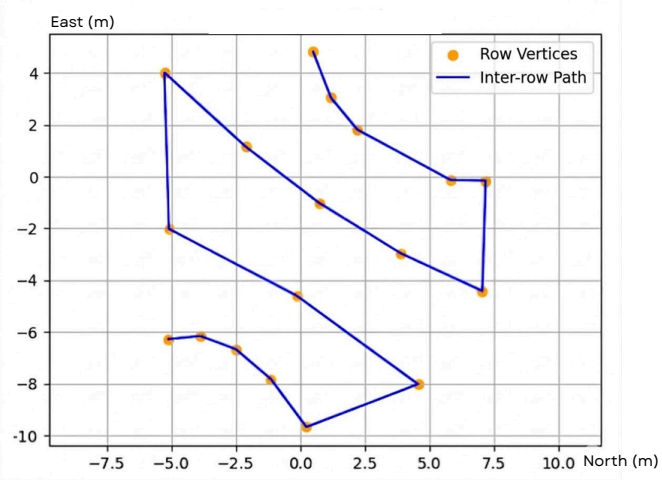


Fig. 5. Inter-row flight path result

traversal order, the vertices are first enclosed within a bounding rectangle defined by the minimum and maximum coordinates. Candidate starting vertices are selected among those located near the rectangle boundaries (within a distance $\epsilon = 0.1\,\mathrm{m}$). From each candidate, a path is constructed by iteratively selecting the nearest unvisited vertex, and the path with the smallest total distance is chosen as the final ordering. This strategy works because vertices within the same Voronoi corridor are significantly closer to each other than to vertices in adjacent rows. To detect transitions between corridors, the average distance between consecutive vertices is computed, and a jump is identified when the distance exceeds a threshold α times this average ($\alpha = 10$ by default).

The resulting paths are then smoothed by fitting a polynomial regression model of degree up to 5 to each corridor segment to reduce abrupt changes in direction. The polynomial degree is selected through cross-validation using the mean squared error as the model-selection criterion. Finally, the trajectory is simplified using the Ramer–Douglas–Peucker algorithm with a tolerance of $0.1\,\mathrm{m}$. The final trajectory is expressed in the local NED coordinate frame and executed by the UAV as a second autonomous flight dedicated to inspection between crop rows. The inspection altitude is a mission parameter that must be set above the canopy height, since no online obstacle avoidance is performed. Although this configuration reduces the risk of collisions with the simulated tree canopies, it does not guarantee collision-free navigation if branches or other obstacles extend into the planned flight corridors and are not represented in the detected geometry. During this flight, the UAV follows the generated waypoints while keeping the camera over the inter-row corridors and capturing additional crop imagery. The potential benefits of this strategy for crop inspection compared with conventional aerial surveys remain a hypothesis that has not been evaluated in this work. Fig. 5 shows an example of the final ordered and smoothed path obtained from the Voronoi-based corridors.

TABLE I
RECONNAISSANCE FLIGHT RESULTS

Metric	Horizontal rows	Oblique rows
Maximum downward error (m)	0.55	0.55
Reconnaissance area (m^2)	542.6	190.2
Avg. reconnaissance time (s)	273.9	114.3
Std. dev. reconnaissance time (s)	3.06	4.4
Coverage cost ($\mathrm{s/m}^2$)	0.50	0.60

F. Architecture

The system is implemented as a modular architecture comprising several ROS 2 nodes responsible for different stages of the pipeline. These nodes handle tasks such as perimeter extraction, reconnaissance path generation, flight control, image acquisition, orthomosaic generation, and row-based path planning. The UAV is controlled by the PX4 flight stack, and communication between PX4 and ROS 2 is provided via Micro XRCE-DDS, enabling real-time exchange of telemetry data and control commands. External processing tasks, such as orthomosaic reconstruction with ODM, are executed as separate processes integrated into the ROS 2 workflow. This modular architecture allows each component of the inspection pipeline to operate independently while maintaining synchronised data exchange across the system.

IV. EXPERIMENTS

This section evaluates the proposed UAV-based pipeline in a simulated agricultural environment. The experiments assess whether the system can obtain a usable representation of the crop, generate navigation paths, and execute them under different operating conditions. To evaluate the proposed system, simulation experiments were conducted in Gazebo, where two distinct crop-field environments were modelled: one with horizontal rows and another with oblique rows. Both the reconnaissance and the inter-row flights were performed at $10.0\,\mathrm{m}$ above the takeoff point and a cruise speed of $2.0\,\mathrm{m/s}$, with a depth camera of $640 \times 480\,\mathrm{px}$ resolution and a 73° horizontal field of view. To assess the UAV's behavioural consistency, 50 trials were conducted for each scenario.

A. Reconnaissance Flight

During the reconnaissance flight, the UAV maintained a stable altitude ranging between $9.77\,\mathrm{m}$ and $10.26\,\mathrm{m}$ (mean $\pm$ one standard deviation), with a maximum downward error of $0.55\,\mathrm{m}$. The spacing between consecutive flight lines showed only minor variations, resulting in a consistent overlap between captured images.

To cover the $542.6\,\mathrm{m}^2$ field with horizontal rows, the average execution time was $273.9 \pm 3.06\,\mathrm{s}$, whereas the $190.2\,\mathrm{m}^2$ field with oblique rows required $114.3 \pm 4.4\,\mathrm{s}$. This corresponds to an approximate coverage cost of 0.50 and $0.60\,\mathrm{s/m}^2$ respectively. Table I summarises the reconnaissance flight results obtained in both scenarios.

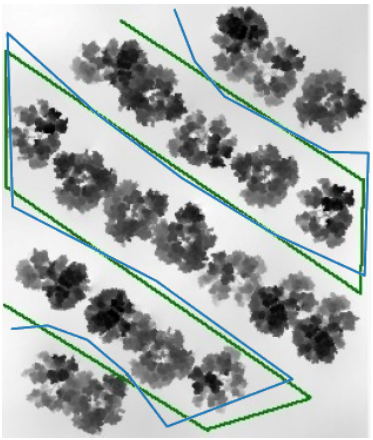


Fig. 6. Manually defined reference (green) and inter-row flight path (blue)

B. Height Map and Inter-Row Path

After the reconnaissance flight, the captured images were processed with ODM to generate a georeferenced height map. In both scenarios, the generated maps preserved the crop structure and provided sufficient information to detect rows and generate paths. Additional tests revealed limitations in height map generation for sparsely distributed low-canopy crops.

To evaluate the quality of the inter-row flight path, the generated path was compared with a manually defined reference path using the Structural Similarity Index (SSIM), which measures similarity across luminance, contrast, and structural information. We selected this metric because the objective was not only to compare pixel-wise overlap, but also to evaluate whether the generated path preserved the overall geometric structure of the reference trajectory [13]. A minimum SSIM threshold of 0.75 was selected empirically based on preliminary tests. Values above this threshold were associated with complete corridor coverage, whereas lower values were observed when segments were missing or misplaced. The horizontal-row and oblique-row scenarios achieved SSIM values of 0.875 and 0.779, respectively, both exceeding the threshold. These results indicate that the generated paths retained structural similarity to the reference paths in the evaluated scenarios.

The lower similarity observed in the oblique-row scenario can be attributed to increased irregularities in the detected canopy structure, as the path generation relies on tree canopy contours. These variations introduce local deviations in the Voronoi-based paths, which are subsequently smoothed through polynomial regression, potentially leading to greater divergence from the manually defined trajectory.

Fig. 6 compares the manually defined path with the automatically generated path for the oblique-row scenario.

C. Inter-Row Inspection Flight

The UAV follows the generated inter-row path at 10 m above the takeoff point and above the canopy of the simulated trees, with the camera centred over the inter-row corridors. The executed paths remained close to the planned ones, indicating that they are not only geometrically valid but also feasible under the simulated conditions.

TABLE II
INTER-ROW PATH AND FLIGHT RESULTS

Metric	Horizontal rows	Oblique rows
Path SSIM	0.875	0.779
Inter-row path length (m)	74.56	57.06
Avg. inter-row flight time (s)	61.59	58.58
Std. dev. inter-row flight time (s)	0.97	3.5

Fig. 7 shows the planned inter-row path with the trajectories executed in the oblique-row scenario, which represents the more geometrically challenging configuration.

In the horizontal-row scenario, the planned path length was $74.56\,\mathrm{m}$, with an average execution time of $61.59 \pm 0.97\,\mathrm{s}$. In the oblique-row scenario, the path length was $57.06\,\mathrm{m}$, with an average execution time of $58.58 \pm 3.5\,\mathrm{s}$. Despite the difference in path length, the execution times are relatively similar. This behaviour can be explained by the greater number of vertices in the oblique-row trajectory, which forces the UAV to decelerate more frequently in order to maintain trajectory accuracy.

Table II summarises the inter-row flight results obtained in both scenarios.

Additional simulations under constant wind of approximately 10.39 m/s in the oblique-row scenario showed that the altitude stability was maintained, with a maximum downward error of $0.49\,\mathrm{m}$, and the inter-row flight time increased only to $60.10 \pm 1.3\,\mathrm{s}$. Despite the reconnaissance coverage cost rising to $0.94\,\mathrm{s/m^2}$ compared to $0.60\,\mathrm{s/m^2}$ under calm conditions, the generated missions remained executable and structurally consistent, suggesting operational robustness under simulated conditions.

Overall, the results show that the proposed system can consistently integrate reconnaissance flight, height map generation, row detection, inter-row path planning, and autonomous flight path execution in simulation. Although additional validation in real agricultural environments is still required, the experiments support the feasibility of the approach as an

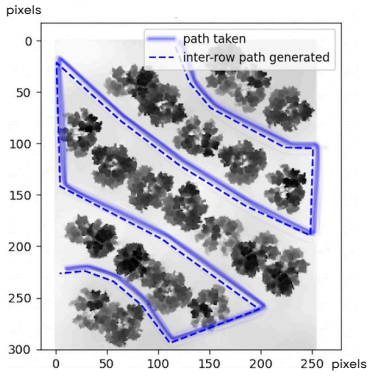


Fig. 7. Planned and executed inter-row path

autonomous data-acquisition and navigation framework for future crop-inspection tasks.

V. Conclusions and Future Work

This paper presented an autonomous UAV-based system for crop inspection and terrain-adaptive path planning in row-structured agricultural environments. The proposed approach combines a reconnaissance flight, height map generation from RGBD imagery, crop-row detection, and inter-row trajectory generation based on a GVD. In this way, the system can transform an initial aerial survey into a subsequent autonomous flight specifically designed to navigate between crop rows.

The simulation results show that the proposed pipeline consistently generates feasible, structured flight paths for both horizontal and oblique row layouts. The reconnaissance stage provided stable flight behaviour and suitable image acquisition for subsequent processing, while the path-generation stage produced trajectories that remained close to manually defined reference routes. In addition, the inter-row flight experiments showed that these paths can be executed reliably under the evaluated simulated conditions.

In general, the results support the feasibility of the proposed system as an autonomous data-acquisition and navigation framework for agricultural inspection. At the same time, the study highlights current limitations, particularly the dependence on row-structured crops with sufficient height contrast and the fact that validation was performed only in simulation. In particular, row detection was evaluated on simulated canopies, and its performance may degrade with real foliage, varying illumination and positioning errors. Furthermore, although the proposed trajectories are intended to support subsequent weed-detection tasks, this study does not evaluate weed-detection performance or establish whether the planned flight paths improve detection outcomes. These aspects require further investigation using real-world imagery and appropriate weed-detection algorithms.

Building upon the simulation results, future research will focus on transitioning the system toward full-scale field deployment through the following key avenues:

Empirical Validation: Extensive real-field experiments will be conducted across diverse crop types, terrain topologies, and weather conditions. These trials are essential for assessing the system's generalisation and for bridging the "sim-to-real" gap.
Enhanced Human-Robot Interaction: To improve usability for non-expert operators, we aim to replace the initial manual reconnaissance flight with more intuitive interface methods. This includes automated mission planning or gesture-based controls to lower the barrier for agricultural deployment.
Robust Perception Pipelines: Future iterations will refine height map generation techniques. We aim to develop more flexible pipelines capable of maintaining accuracy under challenging lighting conditions or dense, irregular crop canopies.
Obstacle Avoidance and Safety: We plan to integrate active collision-free navigation modules. This skill is critical for ensuring the safety of the platform and surrounding infrastructure during autonomous low-altitude flights in unstructured environments.
Vision-Based Weed Detection: Following real-world validation of the navigation pipeline, a vision-based weed-detection module will be integrated and evaluated using annotated agricultural imagery. Future experiments will investigate whether the proposed flight trajectories provide suitable image coverage and improve weed-detection performance compared with alternative image-acquisition strategies.

References

[1] M. Abdulsalam, K. Ahiska, and N. Aouf, "A novel uav-integrated deep network detection and relative position estimation approach for weeds," *Proceedings of the Institution of Mechanical Engineers, Part G*, vol. 237, no. 10, pp. 2211–2227, 2023. [Online]. Available: https://doi.org/10.1177/09544100221150284

[2] A. Q. Dellepiane, T. E. Kaspary, and M. A. G. Latasa, "Capítulo 6: Malezas en sistemas ganaderos y agrícolas," Instituto Nacional de Investigación Agropecuaria (INIA) Uruguay, Informe técnico Aportes científicos y tecnológicos, Capítulo 6, 2023, en: García-Inza, G.P.; Paruelo, J.M.; Zoppolo, R. (Eds.), *Aportes científicos y tecnológicos del INIA-Uruguay a las trayectorias agroecológicas*. [Online]. Available: https://www.inia.uy/sites/default/files/publications/2024-10/Capitulo6Malezasensistemasganaderosyagricolas.pdf

[3] U. Bhandari and A. Etienne, "Precision weed detection using uavs and deep learning: Models, paradigms, and challenges," *Smart Agricultural Technology*, vol. 13, p. 101656, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772375525008871

[4] O. G. Ajayi, J. Ashi, and B. Guda, "Performance evaluation of yolo v5 model for automatic crop and weed classification on uav images," *Smart Agricultural Technology*, vol. 5, p. 100231, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2772375523000618

[5] L. Sandoval-Pillajo, I. García-Santillán, M. Pusdá-Chulde, and A. Giret, "Weed detection based on deep learning from uav imagery: A review," *Smart Agricultural Technology*, vol. 12, p. 101147, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S277237552500379X

[6] S. Gokool, M. Mahomed, A. Clulow, M. Sibanda, R. Kunz, V. Naiken, and T. Mabhaudhi, "Exploring the potential of remote sensing to facilitate integrated weed management in smallholder farms: A scoping review," *Drones*, vol. 8, no. 3, 2024. [Online]. Available: https://www.mdpi.com/2504-446X/8/3/81

[7] S. Meesaragandla, M. P. Jagtap, N. Khatri, H. Madan, and A. A. Vadduri, "Herbicide spraying and weed identification using drone technology in modern farms: A comprehensive review," *Results in Engineering*, vol. 21, p. 101870, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2590123024001233

[8] M. Krestenitis, E. K. Raptis, A. C. Kapoutsis, K. Ioannidis, E. B. Kosmatopoulos, and S. Vrochidis, "Overcome the fear of missing out: Active sensing uav scanning for precision agriculture," *Robotics and Autonomous Systems*, vol. 172, p. 104581, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0921889023002208

[9] F. N. OrtataŞ, H. UlutaŞ, M. E. Şahİn, and F. Çİftcİ, "Autonomous mapping and spraying in precision agriculture using unmanned ground vehicles," in *2023 Innovations in Intelligent Systems and Applications Conference (ASYU)*, 2023, pp. 1–5.

[10] E. K. Raptis, K. Englezos, O. Kypris, M. Krestenitis, A. C. Kapoutsis, K. Ioannidis, S. Vrochidis, and E. B. Kosmatopoulos, "Cofly: An automated, ai-based open-source platform for uav precision agriculture applications," *SoftwareX*, vol. 23, p. 101414, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352711023001103

[11] Z. Ma, X. Wang, X. Chen, B. Hu, and J. Li, "Advances in crop row detection for agricultural robots: Methods, performance indicators, and scene adaptability," *Agriculture*, vol. 15, no. 20, 2025. [Online]. Available: https://www.mdpi.com/2077-0472/15/20/2151

[12] U. Ramer, "An iterative procedure for the polygonal approximation of plane curves," *Computer Graphics and Image Processing*, vol. 1, no. 3, pp. 244–256, 1972.

[13] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.